

HUY TRAN

STAFF SOFTWARE ENGINEER | APPLIED AI

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Staff Software Engineer with 10+ years of full-stack and systems engineering experience, specializing in production Applied AI, multi-agent architectures, and high-reliability financial systems. Proven track record leading the architecture of enterprise-grade invoice extraction pipelines, agentic evaluation harnesses, and distributed backend infrastructure in corporate spend management. High-agency builder who bridges low-level model inference optimization, synthetic dataset curation, and scalable microservices to ship mission-critical AI products.

EXPERIENCE

Navan

Palo Alto, CA

Staff Software Engineer

Apr 2026 – Present

Senior Software Engineer

Feb 2021 – Apr 2026

- Architected multi-agent LLM systems for the Commerce AI team, automating high-volume invoice processing, spend policy verification, and payment discrepancy resolution, slashing manual review overhead by 40% across corporate workflows.
- Designed an in-house LLM evaluation harness featuring multi-metric benchmarking suites and LLM-as-a-judge scoring, shortening prompt and model regression test cycles from days to under 15 minutes while continuously curbing hallucinations.
- Shipped production multi-modal agents, including voice-enabled hotel check-in agents and autonomous expense reconciliation bots embedded directly into core booking and travel flows.
- Built distributed, event-driven streaming microservices supporting mission-critical payment/invoice processing, travel booking, and expense management platforms using Java, Kotlin, Go, and Apache Kafka.

Synchrony Financial

Mountain View, CA

Senior Software Engineer

Feb 2018 – Jan 2021

- Led frontend architecture and checkout API integrations for GiftNow, a high-throughput B2B2C digital gifting platform serving tier-1 national retailers, maintaining 99.99% availability during peak multi-million dollar holiday spikes.
- Spearheaded migration from legacy Backbone.js applications to a modular React and Node.js micro-frontend architecture, decoupling shared codebases to enable multiple engineering pods to deploy independently.
- Optimized critical rendering paths and checkout interaction pipelines, driving user conversion gains while implementing automated telemetry and managing production incident response.

CareSkore

Mountain View, CA

Software Engineer

Aug 2015 – Nov 2017

- Engineered clinical workflow dashboards surfacing machine-learning-driven patient readmission risk factors directly to physicians, alongside a secure, HIPAA-compliant WebRTC consultation platform.
- Built a high-throughput D3.js and TypeScript visualization engine capable of rendering complex multi-chart clinical telemetry streams simultaneously with zero UI thread degradation.

Axon Active Inc

Da Nang, Vietnam

Full-Stack Software Engineer

Nov 2013 – Jul 2015

- Engineered secure mobile banking platforms, digital payment processors, and scalable media streaming microservices for European enterprise clients.

PROJECTS & OPEN SOURCE

- **invoice-classifier**: Trained and open-sourced a domain-specific document classifier optimized for on-device automated invoice and receipt layout categorization, facilitating high-speed metadata routing across financial processing pipelines.
- **pintora-coder-7b**: Fine-tuned a 7B parameter open-source LLM specialized in generating diagram DSL syntax, curating structured training sets to achieve high compilation pass rates for natural-language-to-code diagram generation.
- **Qwen3.6-27B-pure-GGUF**: Engineered low-bit GGUF quantizations for 27B parameter models, benchmarking perplexity and VRAM trade-offs to maximize tokens/sec throughput during local execution via llama.cpp.

TECHNICAL SKILLS

Applied AI & Systems: Multi-Agent Systems, Constrained Decoding (Grammars/JSON), LLM-as-a-Judge Evals, Fine-Tuning (LoRA/QLoRA), RAG, Latency & Token Optimization, LangGraph.

Languages & Core: Python, TypeScript, JavaScript, Go, Rust, Java, Kotlin, C, SQL.

Backend & Infrastructure: Node.js, FastAPI, Spring Boot, PostgreSQL, Redis, Apache Kafka, AWS (ECS, S3, Lambda), Docker, GraphQL.

Frontend: React, Next.js, Web Performance Optimization, TailwindCSS, Micro-Frontends.

EDUCATION

Bachelor of Science in Computer Science — Da Nang University of Education

2009 – 2013